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Carbonic anhydrase traced by cadmium isotope fractionations in Archean to Holocene stromatolitic carbonates



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Stromatolites record interactions of the biosphere, atmosphere, and hydrosphere. The evolution of C-fixation via diverse metabolisms is linked to metal availability for enzymes such as carbonic anhydrase. Stable Cd isotopes fractionation during uptake into carbonic anhydrase by modern phototrophs is readily recorded in authigenic carbonates. Here we test whether this proxy can be expanded to trace carbonic anhydrase in the microbial rock record. We analyzed Cd isotopic compositions in stromatolite occurrences of Western Australia, spanning from the Archean (~ 3.35 Ga Strelley Pool and ~2.72 Ga Tumbiana formations) to the Holocene Hamelin Pool. We record fractionated Cd isotopic compositions correlating with macro and micro-nutrients P, Cd and Zn, suggesting enzymatic metal-cycling in the Paleoarchean. Meanwhile, Rayleigh-type Cd isotope fractionation indicates its use as a co-factor in carbonic anhydrase from the Meso- to Neoarchean onwards. Our study offers new insights into the evolution of the metallome and its role facilitating early microbial metabolisms.

One potential birthplace of early life is shallow aqueous systems, primarily warmed by hydrothermal activity¹. Here, hot, acidic fluids aided the leaching of bioessential metals from an early protocrust, facilitating catalytic processes in early microbes^{2,3}. However, the nuances of the origin of life, especially its energy sources and interaction with metals, also known as the metallome^{4–6}, remain enigmatic.

Methane, which results from the reduction of CO₂ by H₂, may represent the first energy source that chemo-litho-autotrophic archaea could have harnessed for energy conservation⁷. Later in the evolutionary path, heterotrophic bacteria and archaea utilized anaerobic respiration to exploit energy by reducing iron, sulfate, and nitrogen as electron donors; however, the exact onset of these metabolisms in the Archean is still unknown. The evolution of the most drastic innovation in microbial metabolism, the reduction of CO₂ via light, known as photosynthesis, has been speculated to have evolved in the early Archean based on molecular clock studies⁸. The evolution of aerobic photosynthesis, the metabolism that creates free O₂ as a byproduct, is marked by hard geochemical evidence of whiffs of oxygen predating the Great Oxygenation Event (GOE) that starts at around 2.4 billion years ago⁹. One argument for early Archean oxygenation arises from the abundance of stromatolites, i.e., layered bio-sedimentary rocks that form by the incidental

binding and trapping of minerals onto extrapolymeric substances (EPS) of microbial mats¹⁰. The first stromatolites occur in the rock record of the 3.48 Ga Dresser Formation (Western Australia)^{11,12}. Some Archean stromatolites resemble the morphologies of modern oxygenated microbial habitats¹³ and serve as unique geochemical archives to indicate the presence of free O₂ (produced by aerobic photosynthesis), as reported through the occurrence of oxide minerals¹⁴ and redox-sensitive isotope proxies^{9,15}. However, a smoking-gun method for tracing phototrophic metabolisms in early microbial life has yet to be presented. Such demand intrigues bio-geochemists targeting the interdisciplinary fields of geosciences and calls for new geochemical proxies that are crucial to understand metallome evolution in microbial habitats through deep-time.

One of the key metal-enzymes in the microbial world is carbonic anhydrase (CA), which plays an essential role in the carbon cycle of aqueous ecosystems by regulating the balance of CO₂ and HCO₃[–] for assimilation into biomass. In modern oceans, CA is crucial for carbon fixation in aerobic phototrophic primary producers¹⁶, but it is also a common enzyme found in higher heterotrophic organisms¹⁷. The function of CA regulates bicarbonate and carbon dioxide and is especially critical in areas of fast-varying chemical gradients¹⁸.

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One such environment is a microbial mat, where diurnal fluctuation in light penetration causes strong variations in gas exchange between variable microbes (using distinct metabolisms) and ambient pore fluids on a millimeter scale¹⁹. During the daytime, aerobic phototrophs in the uppermost greenish layer of a microbial mat produce excess O₂, which may be partly consumed by the oxidation of organic matter. Below, a purple subsurface layer hosts anaerobic phototrophs (e.g., purple and green sulfur bacteria) that receive residual penetrating light but do not produce O₂. At depth, chemo-litho-autotrophs (e.g., bacterial sulfate reducing bacteria and methanogenic archaea) thrive under anoxic conditions and release free H₂S and CH₄²⁰. While the phototrophic microbes in the upper layers create favorable conditions for authigenic carbonate precipitation, fermentation in the deeper layers causes partial carbonate dissolution, leading to steep chemical gradients and necessitating efficient carbon utilization across diverse microbial communities¹⁹. In the light-prone layers, CA enables phototrophic microbes to modulate the availability of dissolved inorganic carbon (DIC) via the Calvin-Benson Cycle, facilitating a wide range of metabolic processes. In detail, CA in phototrophs catalyzes the conversion of bicarbonate (HCO₃⁻) into CO₂, which can then be utilized by the RuBisCO enzyme in the synthesis of dissolved organic carbon (DOC). In deeper, light-deprived layers, heterotrophs decompose DOC, thereby releasing CO₂ and bicarbonate. Carbonic anhydrase helps a variety of heterotrophic archaea and bacteria counteract acidification, thereby balancing environmental pH and intracellular carbon homeostasis²¹.

Despite its crucial role in carbon fixation and microbial ecology, the exact evolutionary onset of CA is unclear, but it has been speculated to date back to the Archean²¹. Yet, the rock record provides no evidence for the earliest occurrence of CA to date. Cadmium (Cd), a heavy metal with bio-essential functions in the modern ocean, offers such a tracer. In the modern surface ocean, phototrophic primary producers utilize Cd, as evidenced by the remarkable linear correlation between dissolved Cd and macronutrient phosphorus (P) concentrations²². Critically, Cd isotope fractionation via phototrophic primary producers has been reported from culture experiments²³. The findings are linked to the unique cambialistic nature of CA in phototrophic organisms, which means it can utilize Cd as a catalytic co-factor when zinc, its usual catalytic metal, is scarce. This biological uptake of Cd by marine phytoplankton^{24–26} causes nutrient-like distribution patterns in the modern oceans, characterized by surface depletion and depth-dependent regeneration^{27,28}. We propose that combined Cd concentrations and isotopic compositions can be captured in lithified microbial mats (stromatolites)^{29,30}, offering a direct tracer of CA activity. In this study, we apply Cd isotopic fractionation to stromatolitic carbonates from three key microbial habitats to trace the activity of this cambialistic CA through deep time: (i) the Holocene Hamelin Pool in the hypersaline Shark Bay lagoon, (ii) the Archean ~2.72 Ga-old lacustrine Tumbiana Fm., and (iii) the ~3.35 Ga shallow marine Strelley Pool Fm. These locations offer a rare window into three different (paleo)environmental conditions and open the pathway to trace deep-time geo-microbiological metal cycling. By combining Cd isotopic analyses with bio-essential metal concentrations, we aim to trace the activity of Cd-CA and its potential role in carbon cycling and stromatolitic carbonate precipitation through deep-time. We propose Cd as a proxy for exploring the metabolic adaptations of early microbial ecosystems and offer insights into the co-evolution of the metallome and metabolic innovations in the Archean.

Western Australia showcases some of the best-preserved stromatolite occurrences, offering an ideal natural laboratory to study the evolution of microbial metabolisms and the metallome in some of Earth's oldest lithified microbial habitats, specifically in the Archean Pilbara Craton²⁹ and its Holocene analogues at Hamelin Pool in the Shark Bay (Supplementary Fig. 1). The Archean sedimentary Dresser, Strelley Pool and Tumbiana formations (Fm.) deposited from 3.5 Ga onwards on the Pilbara Craton record well-preserved stromatolites and microfossils, providing crucial

evidence of early microbial life^{30,31}. The exceptional preservation of these geological archives (<greenschist facies) compared to their geological age makes these formations crucial sites for understanding the origins and development of early life on Earth. We studied Archean stromatolites from two key formations within the Pilbara Craton, namely the ~3.35 Ga Strelley Pool and the ~2.72 Ga Tumbiana Fms, and compared their geochemical composition to sub-recent analogues from the Hamelin Pool location in the hypersaline lagoon of Shark Bay. The geological settings of the three sample locations (Supplementary Fig. 1), along with their corresponding geomicrobiological contexts, are described in detail below.

The Strelley Pool Fm. (SPF) encompasses meta-sedimentary units with a thickness of up to 400 m and a lateral extent of about 200 km in the East Pilbara Terrain (Fig. 1). It was deposited above a regional unconformity at the top of the 3.43-billion-year-old Panorama Fm., noted as the oldest known hiatus in Earth's history^{30,32}. The SPF is well-preserved (maximum metamorphic overprint at lower greenschist facies⁴⁰), comprising characteristic several-meter-thick stromatolitic and crystal-fan dolomitic carbonates, as well as cherts that preserve seawater-like elemental signatures^{33,34}. The SPF represents one of the oldest known carbonate platforms with a continuous depositional history during overall

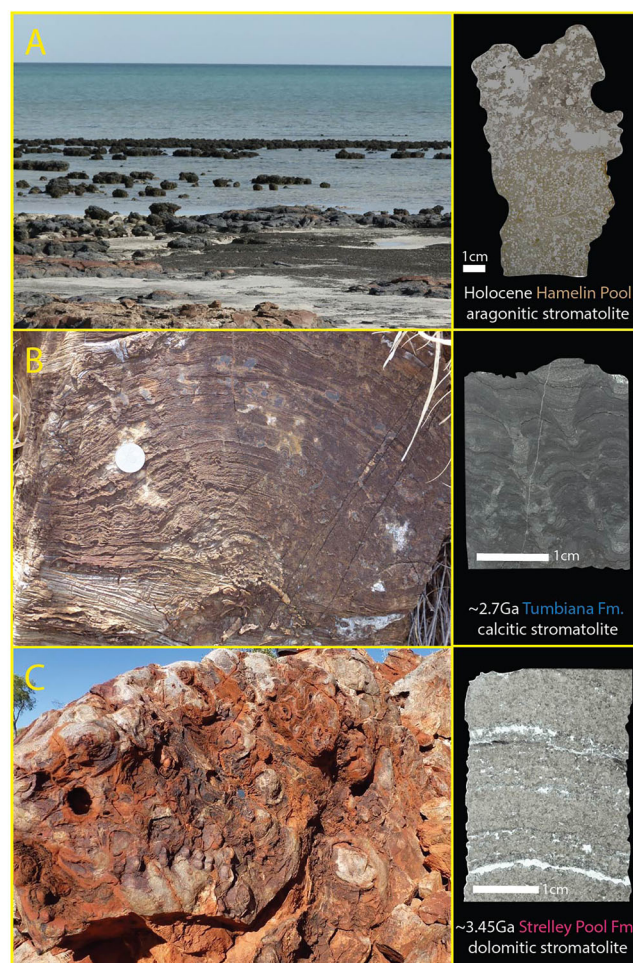


Fig. 1 | Sample locations and thin sections from representative hand specimens. **A** Southern coast of Hamelin Pool, Shark Bay, Western Australia, showing sub-recent inter-tidal stromatolite domes at low tide and near-shore microbial mats with partly lithified crusts; stromatolitic carbonates are primarily aragonitic and show columnar to domal structures on meso- to macro scales. **B** 2.7 Ga Tumbiana Fm. stromatolites in the Fortescue Basin are primarily calcitic and exhibit stratiform to domal macrostructures with microscale columns. **C** Circa 3.35 Ga Strelley Pool Fm. stromatolites on the Pilbara Craton are dolomitic and exhibit domal micro- to macrostructures with abundant laminoid chert fenestrae.

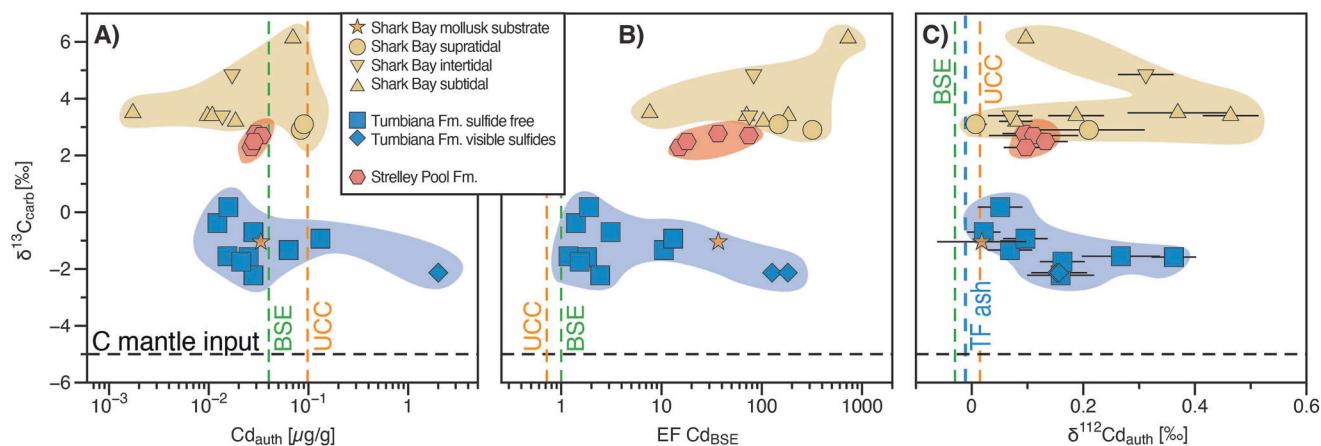


Fig. 2 | Cd concentrations and isotopic compositions in the three studied stromatolite environments relative to their respective C isotopic compositions.

A Authigenic Cd concentrations overlap at variable C isotopic compositions as well as Upper Continental Crust (UCC) and Bulk Silicate Earth (BSE) values (**B**) lower

BSE-normalized Cd enrichment factors at TF relative to SPF and SB. **C** Positive fractionated Cd isotopic compositions above BSE and UCC at all three localities, error bars represent 2SE.

transgression, and diverse stromatolite morphologies representing variable environmental conditions on this ancient marine platform^{31,35}. In-situ S isotope analyses of preserved kerogen and sulfide grains of the stromatolites indicate active bacterial sulfate reduction (BSR) in microbial environments of the Paleoproterozoic³⁶, while light C isotope signatures in bulk organic matter (−35‰) argue for phototrophy³⁷. Stromatolitic samples for this study are aliquots of those geochemically characterized in detail before³³.

The sedimentary Tumbiana Fm. (TF) is a part of the Neoarchean Fortescue Group (2.78–2.63 Ga) (Fig. 1). It is divided into two members, the Mingah and the Meentheena members. The age of the Tumbiana Fm. has been determined through U–Pb zircon dating of bracketing lavas (2741 ± 3 Ma³⁸ and 2717 ± 2 Ma³⁹; and detrital zircons found in tuffaceous sandstones (2721 ± 4 Ma³⁸). The TF has undergone low-grade regional metamorphism, making it an ideal site for studying Neoarchean ecosystems. Stromatolitic (low-Mg calcite) carbonates are found at various levels within the Meentheena Member that represents a lacustrine or a very shallow lagoonal depositional environment dominated by freshwater influx (i.e., absence of characteristics such as current (asymmetric) ripples and tidal channels, flat shale normalized Rare Earth Element patterns)^{40,41}. Stable organic carbon isotope compositions $\delta^{13}\text{C}_{\text{org}}$ indicate that eubacteria and archaea thrived in the microbial habitats⁴². Exceptionally low $\delta^{13}\text{C}_{\text{org}}$ (−40 to −50 ‰) in association with isotopically light $\delta^{13}\text{C}_{\text{carb}}$ (0 to −4‰) in stromatolitic carbonates suggests that methanogens recycled carbon that was previously fixed by phototrophs under highly alkaline conditions^{43,44}. Further, the presence of BSR in the microbial paleo-environment of the TF is corroborated by nanopyrites within pockets of organic matter in the stromatolites⁴⁵.

At Shark Bay, stromatolites serve as modern analogues for microbial carbonate formation dominated by oxygenic phototrophs^{46–48}. Hamelin Pool, a lagoonal-like embayment cut from the open ocean by the Faure sill, hosts a wide variety of living microbial mats and stromatolite morphologies that grow in supra-, intra-, and subtidal environments and are surrounded by tidal flats^{47–49}. Here, carapont formation is tightly linked to microbial activity⁴⁷. In contrast, stromatolite morphologies are not exclusively controlled by microbial communities (i.e., coccoids vs. filaments) but, more importantly, by physicochemical environmental conditions (i.e., water depth, wave strength, geochemical gradients)^{46,48,50}. Carbonate growth rates have been estimated to be <0.4 mm/year based on ¹⁴C analyses⁵¹. The consortia of organisms described from the Shark Bay microbial mats include proto- and cyanobacteria⁵², algal eukaryotes⁵³, and viruses (amongst others)⁵⁴. Samples for this study are aliquots of those geochemically characterized in detail before⁴⁹.

Results and discussion

Authigenic Cd concentrations and C–Cd isotopic compositions in Archean to Holocene W. Australian microbialites

Authigenic Cd concentrations in mild acetic acid carbonate leachates were calculated via normalizing the Cd/Al ratio of individual samples to the Cd/Al ratio of the bulk silicate Earth (BSE) and range from 26.2 to 33.7 ng/g in SPF, 13.5–2004 ng/g in TF, and 1.7–90.6 ng/g in SB, respectively (Eqs. 2–4). Bulk digestion of an associated volcanic tuff at TF yielded authigenic Cd concentrations of ~9 μg/g. An acetic leachate of the mollusk bedrock at SB yields 33.9 ng/g Cd. Carbonate-associated P (CAP) concentrations range from 24 to 199 μg/g, with SPF falling between 157 and 199 μg/g, TF between 30.7 and 72.6 μg/g, and SB between 24.1 and 556. μg/g, respectively (Fig. 2A). Carbonate-associated S (CAS) concentrations range from 67.7 to 313 μg/g, while SPF records values from 67.7 to 139 μg/g, TF ranges from 78 to 313 μg/g, respectively. Concentrations at SB were not determined. Enrichment factors of Cd relative to BSE (Cd_{EF} ; see methods Eq. 1) are highest at SB (>100) and lowest in TF (with two samples <1), where the highest absolute variance of Cd_{EF} is also recorded (Fig. 2B).

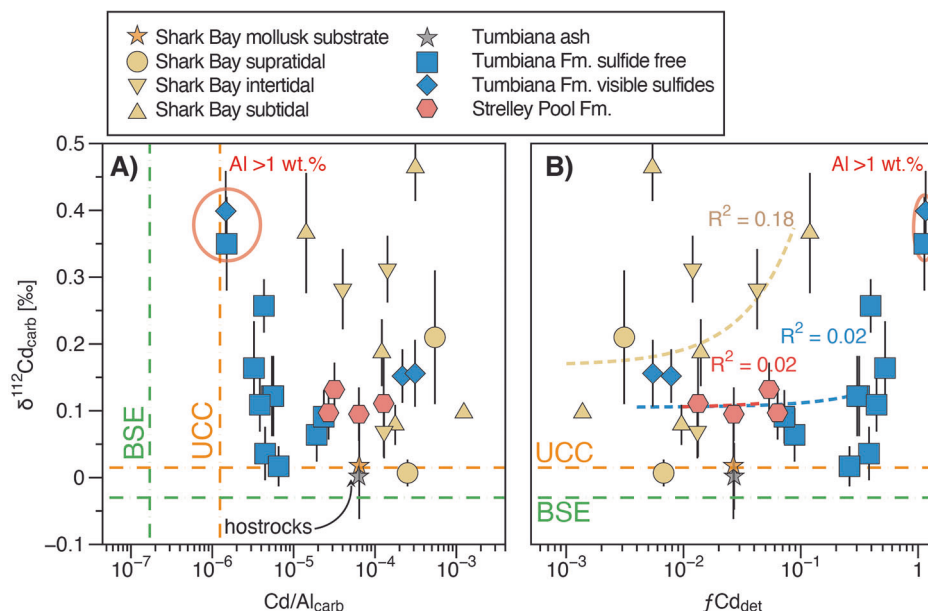
$\delta^{13}\text{C}_{\text{carb}}$ and $\delta^{18}\text{O}_{\text{carb}}$ values of SPF stromatolites show minimal intra-laminae fractionation, ranging from +2.3 to +2.8‰, and −11.6 to −12.3‰, respectively, while TF shows negatively and strongly intra-laminae fractionated $\delta^{13}\text{C}_{\text{carb}}$ values (−2.3 to +0.4‰) and extremely low $\delta^{18}\text{O}_{\text{carb}}$ values down to −16.9‰. Shark Bay stromatolite $\delta^{13}\text{C}_{\text{carb}}$ values are the most positive in the range of +2.9 to +6.1‰ at $\delta^{18}\text{O}_{\text{carb}}$ ranging from +2.8 to +6.0‰. The SB mollusk bedrock records distinct different C–O isotopic values of −1.04 and −0.79‰, respectively (Supplementary Fig. 2).

Authigenic (subscript auth) $\delta^{112}\text{Cd}_{\text{auth}}$, i.e., detrital silicate corrected Cd isotopic compositions derived from authigenic carbonate leaching (methods, Eqs. 2–5), range from +0.1 ± 0.04 to +0.13 ± 0.04‰ at SPF, +0.02 ± 0.03 to +0.45 ± 0.07‰ at TF, and +0.01 ± 0.02 to +0.47 ± 0.05‰ at SB. The mollusk-rich bedrock at SB yields a $\delta^{112}\text{Cd}_{\text{auth}}$ value of 0.02 ± 0.08‰, while the basaltic ash layer at TF yields 0.002 ± 0.05‰, both showing no fractionation from the BSE (Fig. 2C).

Impact of diagenetic overprints and detrital silicate contamination on the stromatolitic carbonates

Various geochemical studies over the past decade have demonstrated the ability of authigenic carbonates to reliably record environmental conditions in microbial habitats during stromatolite formation^{49,55–57}. To test the hypothesis that stromatolitic carbonates can be used as geochemical archives for metallome evolution and specifically the use of CA by phototrophic and heterotrophic microbes as far back as the Paleoproterozoic, we

Fig. 3 | The impact of detrital leaching on the Cd isotopic compositions of stromatolitic carbonates. **A** Despite two samples at TF (with Al con. >1 wt.%, supracrustal Cd/Al ratios argue for authigenic Cd enrichments in stromatolitic carbonates; however, host rock lithologies at TF and SB (grey and orange stars) also record higher ratios than UCC or BSE. **B** Calculated detrital-bound Cd relative to UCC shows no significant correlation with authigenic Cd isotopic compositions, arguing against silicate mineral contamination. Error bars represent 2SE.



employ Cd isotopes in individual layers of stromatolitic carbonates to record Cd fractionation during co-factor uptake or homeostasis^{58,59} in a semi-closed biogeochemical system (the microbial mat and the ambient porewaters from which carbonate minerals precipitate). Our hypothesis states that biological Cd utilization in a semi-closed microbial habitat should follow Rayleigh-type distillation systematics. To interpret the environmental conditions and microbial metabolisms under which Cd has been incorporated into stromatolitic carbonates, the null hypothesis that the observed Cd isotope fractionations and metal enrichments are strictly abiotic must first be rejected. Factors that can lead to abiotic variations in Cd systematics in stromatolitic carbonate leachates are:

- detrital aluminosilicate contamination during carbonate leaching, masking the Cd isotope signal of the carbonate
- diagenetic overprint during early or late fluid-rock interactions with metamorphic or meteoric fluids
- inorganic fractionation processes (e.g., adsorption on Fe-Mn oxy-(hydr)oxides, Cd-sulfide precipitation),
- biological fractionation unrelated to carbonic anhydrase, like non-specific uptake of Cd.

To assess the amount of detrital-bound Cd liberated during leaching (i), we calculated authigenic (subscript auth) Cd concentrations and isotopic compositions in carbonates ($\delta^{112}\text{Cd}_{\text{auth}}$) relative to Cd/Al ratios and isotopic compositions of the BSE⁶⁰. No correlations were observed with authigenic Cd concentrations and almost immobile elements, i.e., elements that are enriched in the silicate crust, such as Al (Fig. 3). The tuffaceous host rock of the TF reveals extreme enrichment in Cd (~9 µg/g) above UCC and BSE values (at ~0.1 and ~0.04 µg/g, respectively) at unfractionated Cd isotope composition ($\delta^{112}\text{Cd}_{\text{auth}} = 0.002 \pm 0.05$). Additionally, two stromatolitic carbonates of TF record elevated Al concentrations above 1 wt.% at relatively low Cd concentrations (15.8 and 22.2 ng/g). If leaching of silicate phases occurred during carbonate digestion, this would have increased the Cd concentration and isotope compositions in the stromatolites and would have shifted the isotope signatures closer to the value of BSE ($-0.03 \pm 0.03\text{‰}$)⁶⁰. The near-zero Cd isotopic composition of the tuff at TF suggests that detrital input would indeed lower Cd isotope values. However, for the high Al stromatolitic carbonates at TF, the opposite is the case, yielding exceptionally heavy $\delta^{112}\text{Cd}_{\text{auth}}$ values (at +0.35 and +0.4‰, respectively). Partly silicified SPF has 96 to 99% authigenic Cd, while TF ranges between 61% and 99% authigenic Cd (excluding two high Al samples (Fig. 3A)), and SB ranges from 91% to 99%, respectively.

After correcting for authigenic Cd (Cd_{auth} ; Eqs. 2–5), and exclusion of samples containing >1 wt. Al, no apparent covariation between $\delta^{112}\text{Cd}_{\text{auth}}$ and the detritus-derived Cd fraction is observed (Fig. 3B). All data discussed below are based on Al (<1 wt.%) - filtered authigenic $\delta^{112}\text{Cd}_{\text{auth}}$ values;

Regarding diagenetic overprint (ii), the stromatolites show variable degrees of lithification and post-diagenetic modification. To analyze the most pristine portions (i.e., least diagenetically altered), we used petrographic pre-screening for post-depositional calcitic veins, followed by layer-specific diamond-coated microdrill sampling. While primary aragonite is still preserved in SB, SPF shows aragonite pseudomorphs in a dolomitic matrix as well as silicification with fenestrae chert structures⁶¹. However, Van Kranendonk et al.³⁴, Allwood et al.⁶¹, and Viehmann et al.³³ have shown that these stromatolitic carbonates still reliably record environmental conditions and ambient fluid chemistry³⁰. Furthermore, Bolhar et al.⁴¹ obtained pristine shale normalized REE patterns in TF stromatolitic calcites. To test whether dolomitization or meteoric fluid overprint had an impact on the Cd isotopic composition of the two Archean stromatolite sample suites, Cd isotopic compositions and Cd concentrations are plotted versus carbonate (subscript carb) Mn/Sr ratios and $\delta^{18}\text{O}_{\text{carb}}$ values (Supplementary Fig. 2). Although Cd is not a fluid-mobile metal, a meteoric overprint could be expected to change isotopic compositions via precipitation of late diagenetic calcitic cements, which record high Mn/Sr ratios and low $\delta^{18}\text{O}_{\text{carb}}$ values. We found no apparent covariation with $\delta^{112}\text{Cd}_{\text{auth}}$ across all three sample locales, suggesting minimal post-depositional alteration of primary Cd signatures in stromatolitic carbonates, thereby corroborating their applicability as geochemical archives for tracing Cd isotope fractionation in Archean and Holocene microbial habitats.

For inorganic fractionation processes (iii), abiotic mechanisms like adsorption onto Mn oxy-(hydr)oxides or inorganic sulfide precipitation cannot explain the systematic positive covariations of $\delta^{112}\text{Cd}$ with phosphate and zinc within microbial laminations. These processes would also cause abundant framboidal sulfide grains in the studied stromatolites, apart from the TF, no petrographic evidence for sulfide formation was found. However, the formation of oxide phases has been shown to exert a potential control on other Archean microbial carbonate systems⁵⁸, however adsorption on these phases would generate only minor isotopic fractionation that tends to reach ~0‰ at longer timescales⁶². Testing the influence of oxide formation at the microscale may rely on covariations between Cd isotopic compositions and Ce anomalies in microbial carbonates, provided that these covariations are preserved rather than inherited from later meteoric diagenesis. Unlike in the Mesoproterozoic Pongola Fm.⁵⁸ no correlations have

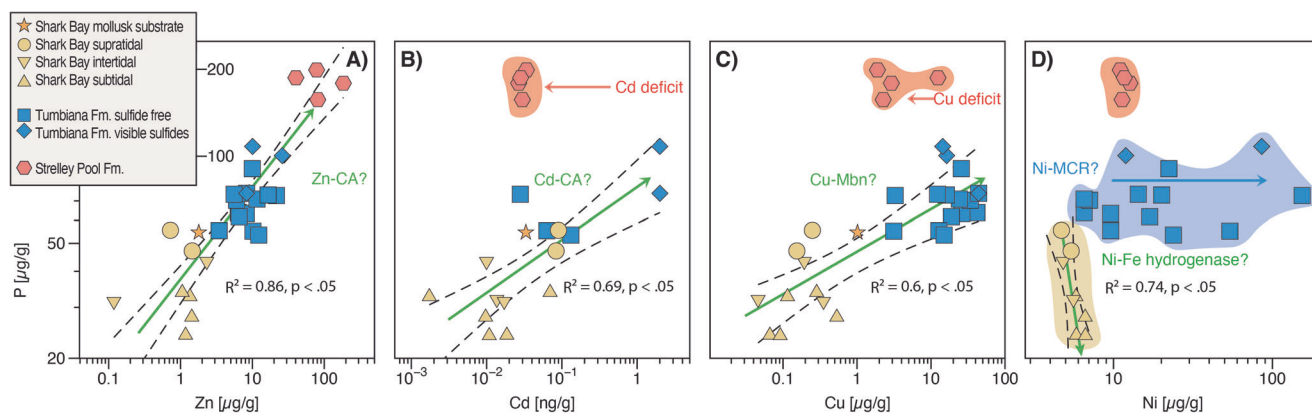


Fig. 4 | Bioavailable metal co-variations in Archean and Holocene stromatolitic carbonates relative to P following the power law. A Macro-nutrient P vs. Zn concentrations in carbonate leachates reveal a positive co-variation through deep time with progressively more depleted values into the Holocene, arguing for an increased biological turnover of Zn in carbonic anhydrase (CA) (**B**, **C**) Co-variation of P vs. Cd or Cu shows an offset to lower concentrations of the trace metals in SPF,

potentially arguing against their use as co-factors (for CA, or Mbn) in Paleoarchean microbial habitats. **D** Distinct enrichments of Ni in Neoproterozoic TF microbial carbonates argue for its use as a co-factor by methanogen archaea (MCR) at that specific locale, while distinct co-variations of P and Ni at the Holocene SB argue for the use of Ni in Ni-Fe hydrogenase used in SRB.

been found between carbonate-hosted Ce anomalies and $\delta^{112}\text{Cd}_{\text{auth}}$, underlining the negligible influence of oxide formation on the observed Cd isotopic compositions (Supplementary Fig. 3).

Finally, nonspecific cellular uptake or bulk homeostasis (iv)⁵⁹ would not generate the systematic biogeochemical metal enrichments in stromatolitic carbonates described below. These processes would produce little to no isotopic fractionation and would not generate the strong, layer-specific positive covariations among $\delta^{112}\text{Cd}$, phosphate, and zinc recorded in the stromatolites. Crucially, they cannot explain the Rayleigh-type distillation trends, which are diagnostic of a finite nutrient reservoir being progressively and fractionally consumed by a specific process. If the metal was primarily treated as a toxin, it would not be systematically depleted in the reservoir. Therefore, the null hypothesis of non-specific biological uptake is rejected.

The combined evidence leads us to reject the null hypothesis that diagenetic, abiotic, or nonspecific uptake processes control Cd isotope fractionation in the studied microbial carbonates. The Cd isotope signatures are best interpreted as recording enzymatic metal processing within stromatolite-forming microbial ecosystems.

Archean microbialites as trace metal sinks

The Archean TF and SPF samples reveal two-to-four times higher concentrations for P, Cd, Zn, Cu, and Ni relative to their Holocene counterparts (Fig. 4). In the primarily anoxic Archean hydrosphere conditions, Fe(III) (oxyhydr)oxides, which are important phosphorus sinks in modern oxygenated oceans⁶³, were likely less effective at scavenging phosphate (PO_4^{3-}). Therefore, larger amounts of dissolved phosphate likely remained in Archean seawater, making it more available for incorporation into carbonates during precipitation. With low biological activity throughout the Archean, P remained primarily in the water column and was readily available for incorporation into carbonate minerals. The sources of P in the Archean were also influenced by intense chemical weathering under a reducing atmosphere, which released greater amounts of phosphorus from continental rocks, making non-ferrous trace metals more limiting than P⁶³. In modern oceans, oxidative weathering processes rapidly adsorb phosphate onto Fe(III) minerals, thereby limiting phosphorus flux to the oceans.

Positive co-variation between Zn and P at all three sample locations suggests that phototrophic nutrient cycling occurred in the Archean. Interestingly, while TF and SB lie on linear trends for all trace metals with respect to P, Cd, and Cu concentrations deviate from this linear correlation trend at SPF (Fig. 4B, C), potentially arguing against their cycling as co-factors for phototrophic carbonic anhydrase and methanotrophic

methanobactins (Mbn) at ~3.35 Ga. Meanwhile, a distinct enrichment in Ni concentration at roughly similar P concentrations at TF relative to the other locales argues for its use in the Wood-Ljungdahl metabolic pathway, like in Methyl CoM reductase (MCR) by methanogen archaea (Fig. 4D)^{7,64}. Distinct P enrichments in idiomorphic acicular minerals observed under electron diffraction spectroscopy (EDS) at TF stromatolite thin sections (Supplementary Fig. 4), and positive co-variations of carbonate-associated P and S concentrations at TF and SPF (Supplementary Fig. 5), may hold some evidence for sulfate-driven anaerobic oxidation of methane and the formation of the Fe-phosphate vivianite. At SB, negative covariations of Ni with P argue for incorporation in Ni-Fe hydrogenase during BSR processes.

Archean stromatolitic carbonates exhibit distinctly higher median Cd concentrations compared to those from the Holocene, which can be attributed to differences in dissolved metal availability under varying redox conditions and/or variable biological activity. One can assume that the absence of a plankton-driven biological pump reduced Cd removal from surface waters, resulting in longer Cd residence times in Archean aqueous systems and higher concentrations available for incorporation into authigenic mineral phases⁶⁵. In contrast, modern oxygenated water bodies have lower surface Cd^{2+} concentrations due to biological uptake, with Cd being sequestered in organic matter and exported to deeper waters, thereby reducing its availability for incorporation into shallow-water carbonates⁶⁶. Additionally, oxidative weathering processes bind Cd to Fe-Mn oxy/hydroxides, further reducing the dissolved continental Cd flux to the modern oceans. Consequently, while Archean carbonates may exhibit higher Cd concentrations, modern carbonates may reflect lower Cd availability in the surface ocean⁶⁷. The Cd sources in the Archean were influenced by weathering under a predominantly reducing atmosphere, likely releasing higher concentrations of Cd^{2+} from a primarily mafic proto-crust into the ocean (SPF) or lakes (TF), thus increasing baseline Cd concentrations. However, the highest Cd EFs were obtained in Holocene stromatolites of SB compared to TF and SPF, suggesting a more effective Cd sink in Holocene microbial carbonates than in Archean microbial carbonates (Fig. 2B).

Holocene microbialite Cd isotope fractionation recorded at Hamelin Pool, Shark Bay – the role of carbonic anhydrase

Carbonates are reliable recorders of Quaternary seawater Cd isotope compositions⁶⁶. By using the salinity-dependent isotope fractionation factor between ambient fluid and carbonates⁶⁸, the Cd isotope signature of authigenic carbonates constrains the $\delta^{112}\text{Cd}_{\text{fluid}}$ values of ancient seawater, thereby unravelling past variations in marine productivity^{69–73}. Calculated

Fig. 5 | Cadmium isotopic composition of microbial carbonates from three different sampling sites in Hamelin Pool lagoon. **A** Negative correlation between Cd isotopic composition and authigenic Cd concentration argues for Rayleigh-type uptake of Cd into biomass and subsequent carbonate formation. **B** Negative correlation with Cd/Zn argues for phototrophic biomass uptake of Cd at variable Zn and P concentrations, as represented by variable symbol size. UCC concentrations and isotopic compositions are from (Pickard et al.⁶⁰). Error bars represent 2SE.

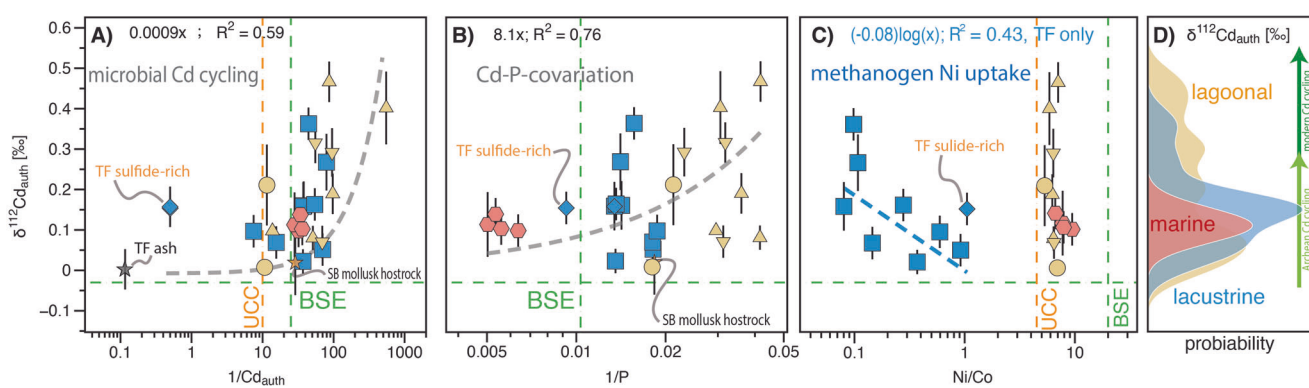
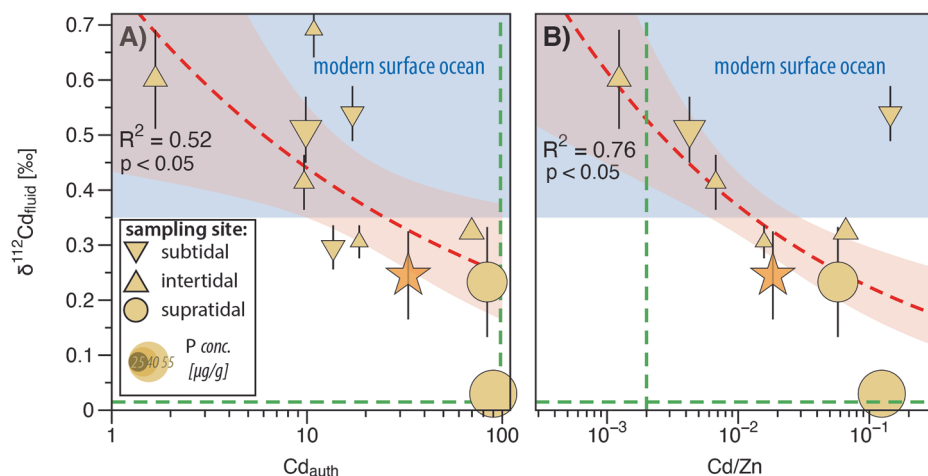


Fig. 6 | Cadmium isotopic compositions in stromatolitic carbonates relative to bioavailable trace and macro nutrients. **A** Positive correlation of $\delta^{112}\text{Cd}_{\text{auth}}$ with $1/\text{Cd}_{\text{auth}}$ arguing for Rayleigh-type Cd fractionation in individual microbial habitats, with the strongest signals in the Holocene lagoonal environment. **B** General positive co-variation of $\delta^{112}\text{Cd}_{\text{auth}}$ with $1/\text{P}$ argues for a link between macro-nutrient P with Cd, similar as in modern productive surface oceans. **C** Unlike the other two localities,

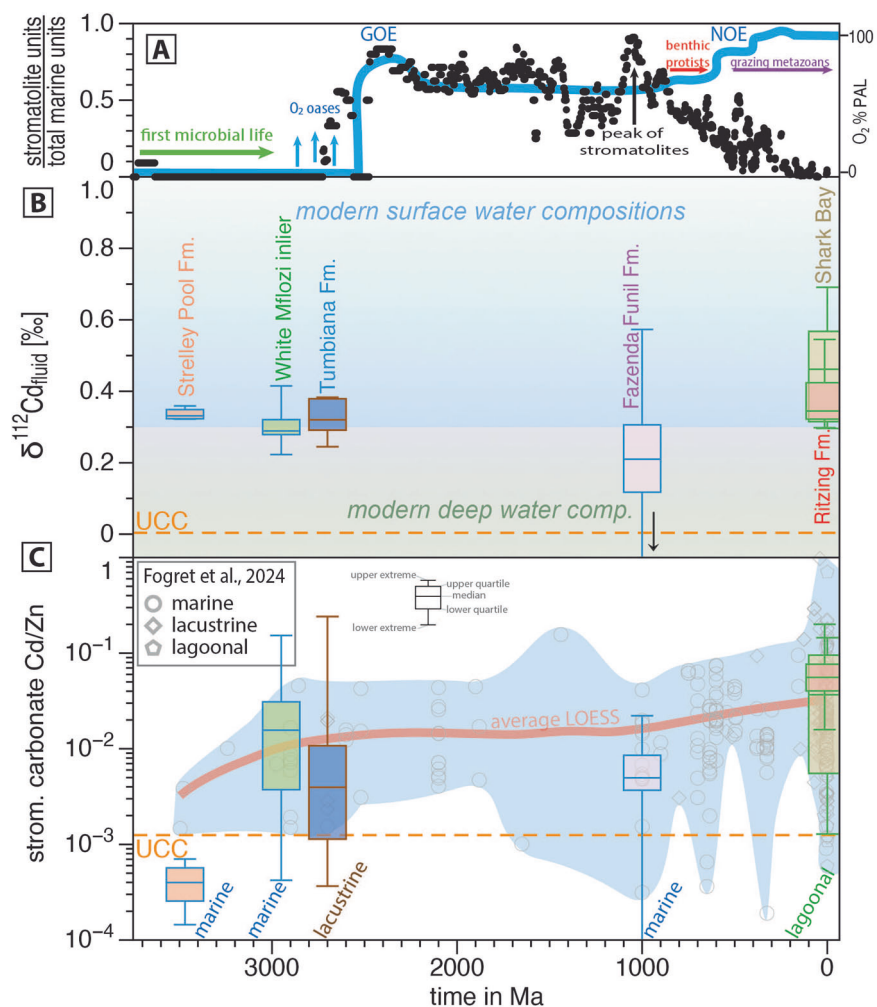
TF shows strong logarithmic co-variation of $\delta^{112}\text{Cd}_{\text{auth}}$ with Ni/Co arguing for Ni uptake as co-factor in methanogen archaea. **D** Probability histogram of $\delta^{112}\text{Cd}_{\text{auth}}$ shows that the heaviest isotopic compositions occur within lagoonal Holocene SB stromatolites, while paleo-marine SPF samples show narrow variation in a small dataset ($n = 4$). Error bars represent 2SE; for the symbol legend, refer to Fig. 4.

$\delta^{112}\text{Cd}_{\text{fluid}}$ values of ambient fluids in equilibrium with the studied stromatolitic carbonates (Fig. 5) overlap with to those of modern surface oceans⁷⁴, with a temporal evolution from the Archean to the highest recorded values in Holocene Hamelin Pool stromatolitic carbonates (Fig. 6). We speculate that the reason for the observed Cd isotope fractionation in paleo-microbial environments is enzymatic Cd uptake by primary producers such as phototrophic bacteria and algae, presumably as a co-factor in the CA enzyme, as reported from culture experiments²³, water and sediment samples^{74–77}. The operational use of Cd as a co-factor in CA is mechanistically linked to the specific environmental conditions of microbial habitats, i.e., restricted, evaporative settings with high salinity, high pH, and likely localized sulfide production within organic-rich mats which would have reduced Zn^{2+} bioavailability through complexation and precipitation⁷⁸. In a Zn-limited niche, Cd^{2+} may have remained relatively more accessible. This environmental pressure likely favors the uptake of Cd-into CA⁷⁶, particularly in key autotrophic organisms like diatoms and proteobacteria present in the community²⁶. The subsequent preferential biological uptake of light Cd isotopes into enzymes would directly explain the observed isotopic fractionation patterns in stromatolitic carbonates.

Microbial carbonates from the Holocene Hamelin Pool have been geochemically characterized in detail by Martin et al.⁴⁹. For this study, we selected stromatolite samples of this study from three locations at different

water depths (supratidal, intertidal, and subtidal) with corresponding morphologies (pustular, colloform, and cerebroid)⁴⁸. At the locations, water was sampled in direct association with the stromatolites, and alkalinity, cation concentration, and salinity have been determined⁴⁹. Samples from subtidal and intertidal environments have salinities roughly twice that of modern open seawater (65.6 and 59.8 PSU, respectively), whereas a microbialite from the shore above the waterline (supratidal) formed under groundwater influx at 3.6 PSU. To calculate the ambient $\delta^{112}\text{Cd}_{\text{fluid}}$ in equilibrium with carbonates, we employed an assumed linear correlation between salinity and the fractionation factor of Cd isotopes between the fluid and the carbonate that precipitated from it⁷³. The calculated $\delta^{112}\text{Cd}_{\text{fluid}}$ values at Hamelin pool show negative correlations with Cd_{auth} and Cd/Zn ratios (Fig. 5), which also correlates with the concentrations of macronutrient P in the stromatolite samples (Fig. 4). This strongly argues for a Rayleigh-type fractionation behavior with consumption of Cd and P as well as light Cd isotope incorporation into carbonates leading to progressively depleted dissolved Cd concentrations in a semiclosed microbial mat system. Supratidal stromatolitic carbonates growing close to a spill site of groundwaters from the Tamala aquifer show high Cd_{auth} concentrations at low salinity, and $\delta^{112}\text{Cd}_{\text{fluid}}$ close to UCC⁶⁰; in contrast, subtidal and intertidal stromatolitic carbonates record isotopically heavier values with $\delta^{112}\text{Cd}_{\text{fluid}}$ values overlapping with highly productive surface seawaters⁶⁷ at low Cd and Cd/Zn ratios (Fig. 5).

Fig. 7 | Cadmium isotopic evolution of ambient fluids in equilibrium with stromatolitic carbonates through time. **A** Relative abundance of stromatolite-bearing sedimentologic units relative to total recorded marine units from the N.-American stratigraphy⁸⁶ indicates a rapid increase in stromatolites short before the Great Oxygenation event (GOE) and again before the Neoproterozoic Oxygenation event (NOE) as shown by the atmospheric O₂ levels relative to present atmospheric levels (PAL)⁸⁵. The decline in stromatolite occurrences coincides with the evolution of higher benthic organisms such as protists and metazoans. **B** Box and whisker plots of absolute Cd isotope variations in fluids of microbial habitats calculated in equilibrium to stromatolitic carbonates from six different (paleo) environments: ~3.45 Ga Strelley Pool Fm. (Australia); ~2.95 Ga White Mfiozi inlier in the Pongola Supergroup (S. Africa)⁵⁸; ~2.72 Ga Tumbiana Fm. (Australia); ~1 Ga Paranoá Group (Brazil)⁸⁷; Miocene Ritzing Fm. (Austria)⁸⁸. A temporal trend to more pronounced heavy Cd isotope compositions can be observed from the Archean to Phanerozoic, while the Proterozoic Fazenda Funil Fm. from the Paranoá Group shows extensive light Cd recycling⁸⁷. **C** Box-and-whisker plots of Cd/Zn ratios in the above-mentioned stromatolitic carbonate localities, plotted on top of a database of metal abundances in microbial carbonates by Fogret et al.⁸⁴, indicating a steady increase in Cd over Zn usage in microbial habitats (red LOESS trend) irrespective of the paleo- environmental depositional conditions, i.e., marine, lacustrine or lagoonal.



The specific microbial community of a living microbial mat in Hamelin Pool has been analyzed via 16 s rRNA analyses⁵², revealing the presence of alpha and gamma protobacteria, cyanobacteria, bacteroidetes, planktomycetes and actinobacteria.

Positive co-variations between macronutrient P and micronutrients Zn, Cu, and Cd argue for their direct use as metallic co-factors in this microbial environment (Fig. 4). While Zn is associated with phototrophic carbonic anhydrases and Cu suggests the presence of methanotrophic methanobactins⁷⁹, the obtained co-variation of P and Cd argues for the use of Cd as a crucial metallic co-factor in microbial habitats. Culture experiments have shown a non-specific uptake by protobacteria and subsequent storage of the heavy metal Cd in cell walls (homeostasis) as a driving factor for Cd isotope fractionation⁵⁹. In the stromatolites of Hamelin Pool, over 50% of the microbial biomass belongs to the group of protobacteria⁵². In contrast, active Cd uptake by eukaryotes may account for kinetic isotope fractionation in microbial communities. For instance, diatoms may use a variant of CA that substitutes Zn for Cd²⁶, which is important because subtidal colloform and cerebroid stromatolites at Hamelin Pool contain abundant diatom assemblages coexisting with cyanobacteria that stabilize the microbial mat via EPS excretion^{48,80}. Thus, the observed heavy Cd isotopic compositions in microbial carbonates of Hamelin Pool may stem from precipitation of a Cd-depleted ambient fluid following the preferential uptake of Cd into Cd-CA of prokaryotes and eukaryotes in microbial mats. The active uptake of Cd in CA is recorded in the negative correlation between δ¹¹²Cd_{fluid} and Cd/Zn ratios at comparable P concentrations (size of the symbols in Fig. 5B) in the Holocene Hamelin Pool stromatolitic carbonates.

Enzymatic Cd isotope fractionation in Archean microbial habitats and implications for the metallome of early phototrophs

Positive co-variations of P and Zn in carbonate leachates of all three sample localities (Fig. 4A) provide strong evidence for the use of the macro- and micronutrients by phototrophs from as early as the Paleoproterozoic. Additionally, this study observed clear positive Cd isotope fractionation above the BSE and UCC background in Archean stromatolitic carbonates from the TF and SPF (Fig. 6) and calculated ambient fluid δ¹¹²Cd_{fluid} compositions at the two paleo-environments overlap with modern productive surface ocean compositions (Fig. 7). To link the observed Cd isotope fractionation to phototrophic producers, and a potential uptake of Cd as co-factor into CA like in the Holocene microbial communities of SB, we first need to rule out fractionation via the other strong sink for light Cd, i.e., Cd-bearing sulfides (CdS)⁸¹. No sulfide minerals are found in thin sections of Holocene SB, yet the two Archean SPF and TF localities contain small amounts of dispersed framboidal and semi-euhedral sulfide grains (Supplementary Fig. 4). While at SPF, these grains are rare and largely dispersed, in the TF stromatolites, two hand-specimens show layer-specific sulfide accumulation (Supplementary Fig. 4), indicating BSR horizons⁴⁵. A strong positive correlation is observed between elevated carbonate-associated S (CAS) and heavier δ¹¹²Cd_{carb} in the TF (R² = 0.91, n = 9; Supplementary Fig. 6), arguing against widespread CdS formation as the primary control. If extensive sulfate reduction was a dominant process at TF, one would expect to see a negative correlation, i.e., high sulfide production would simultaneously draw down the sulfate reservoir (resulting in low CAS values) and sequester light Cd isotopes into sediments (leaving the dissolved pool and carbonates enriched in heavy Cd isotopes)⁸¹. In contrast, high CAS concentrations, coinciding with heavy Cd

Table 1 | Cd isotope fractionation in microbial habitats through deep time and its interpreted context

Eon/Era	Age (Ga)	Formation	Key Cd Isotope Feature ($\delta^{114}/^{110}\text{Cd}$)	Evidence for biological Cd uptake?	Associated Proxies (P, Zn, etc.)	Interpreted Context
Paleoarchean	~3.35	Strelley Pool (this study)	Moderately fractionated	Equivocal	Positive P, Zn correlations, but low Cd/Zn ratio	possibly abiotic-influenced
Mesoarchean	~2.95	Pongola (Hohl et al., 2024)	Fractionated, strong Rayleigh trend	Yes	Bio-intrinsic metal accumulation in stromatolitic layers, high Cd/Zn ratio	Enzymatic fractionation by phototrophic use of CA
Neoproterozoic	~2.72	Tumbiana (this study)	Fractionated	Yes	Positive P, Zn correlations, high Cd/Zn ratio	enzymatic fractionation by phototrophic use of CA, BSR and S-AOM
Mesoproterozoic	~1	Fazenda Funil (Viehmann et al., 2019)	Fractionated, strong Rayleigh trend	Yes	Variable Cd/Zn ratio, redox-sensitive metal vs Cd co-variation	Persistent enzymatic Cd use under nutrient starved conditions (recycling of biomass)
Miocene	~0.014	Ritzing (Hohl et al., 2026)	Fractionated	Yes	High correlation with CNS and Ba nutrient proxies	Strong nutrient fluctuation during the Badenian salinity crisis and associated evaporative events
Holocene	~0	Hamelin Pool (this study)	Near-modern seawater	Calibration point	Strong co-variation with P and Cd/Zn	Modern analogue

isotope compositions, indicate that the sulfate reservoir remained robust and that the Cd isotope signature instead faithfully records increased biological productivity, driving CO_2 drawdown (increasing alkalinity and CAS formation) and high nutrient utilization (causing the shift to heavier $\delta^{112}\text{Cd}$), with sulfide formation being restricted to only some levels within the microbial mat stack (Supplementary Fig. 4). Nano-scale sulfides of individual layers within the TF stromatolites are interpreted to have a microbial origin, specifically linked to BSR and sulfur-dependent anaerobic oxidation of methane (AOM)^{45,82}, a negative logarithmic correlation is observed between Ni/Co ratios (Ni acts as a co-factor essential for methanogenesis) and $\delta^{112}\text{Cd}_{\text{auth}}$ (Fig. 6C). We can only speculate that while BSR played an important role, authigenic sulfides did not considerably incorporate Cd, thus the observed isotopic Cd fractionation of the stromatolitic carbonates (Fig. 6D) resulted from phototrophic metabolism^{42,44}. Indeed, at the TF, positive co-variations of $\delta^{112}\text{Cd}_{\text{auth}}$ with 1/Cd and 1/P (Fig. 6A, B) at variable Cd/Zn ratios (Fig. 7C) argue for the interchangeable use of Zn and Cd as metallic co-factors in CA, resulting in the strong observed isotopic fractionation of Cd in the Neoproterozoic lacustrine Tumbiana ecosystem.

The Strelley Pool paleo-environment, which likely represented open-marine conditions, exhibits both heavy Cd and C isotopic compositions with a distinct positive correlation ($R^2 = 0.57$). At the SPF, microfossil assemblages were discovered in associated black cherts⁸³, which preserve molecular signatures (such as high N/C values and amide groups)⁴⁴, suggesting diverse microbial communities. However, the absence of definitive biomarkers, such as hopanoids or isoprenoids, suggests a lack of phototrophic or autotrophic organisms that typically depend on CA to enhance CO_2 fixation. However, biomarker proxies are poorly preserved in such old stromatolitic carbonates. At SPF, we observe a deviation of Cd concentrations from an expected power law relationship between Cd and P (Fig. 4B). Together with distinct lower Cd/Zn ratios (Fig. 7C) and the lowest recorded $\delta^{112}\text{Cd}_{\text{auth}}$ values in the three studied locales, our data argue against substantial uptake of Cd in a cambialistic CA version during the Paleoproterozoic. Additionally, stable CNS isotope data from the SPF support the dominance of simpler metabolisms, such as methanogenesis and early bacterial sulfate reduction, which do not necessarily require CA⁸². Together, these factors suggest that while CA became important in more evolved carbon fixation pathways during the Meso- to Neoproterozoic, it possesses crucial information on metallome evolution and the first appearance of Cd-CA in the geologic rock record.

Proxy validation and limitations

While the here-presented Cd isotope data robustly demonstrate the utility of Cd isotopes as a tracer of microbial CA activity in microbial carbonates, the

application of this proxy to the rock record requires consideration of several factors. These include preserving primary Cd signatures against late-stage diagenetic overprint and detrital contamination, which must be ruled out using mild leaching methods and rigorous screening of contaminated samples. The influence of early carbonate diagenesis on the Cd isotope proxy is negligible, as Quaternary carbonate platforms reliably record surface-ocean Cd isotopic compositions⁶⁶. The role of organic matter, specifically EPS-related binding of Cd, and its contribution to remineralization remain unresolved. Future research thus needs to systematically address this research gap via laboratory and field studies of living microbial mats. Finally, any deep-time geochemical proxy has gaps in the spatial and temporal data within the geologic record. Our long-term variation curves presented here are intended as an initial reference and should be validated with higher spatial and temporal resolution.

Evolution of Cd cycling in microbial habitats through deep time

This study, combined with previously published data, provides discrete but critical snapshots of Cd isotope systematics in microbial carbonates through deep time (see summary in Table 1). While we do not aim to construct a continuous Cd isotope curve, a goal that will require a much broader geographical and temporal dataset, the currently studied microbialite occurrences allow for a comparative analysis of microbial Cd utilization, especially when comparing our Cd isotope data with a previously published big dataset on stromatolitic carbonate metal abundances through deep time⁸⁴ and its linkage to atmospheric O_2 evolution and stromatolite appearance^{85,86}. The dataset shows a steady rise of Cd/Zn ratios from the Archean to the Holocene in microbial carbonates arguing for the increased uptake of Cd under Zn limited conditions as co-factor for CA (Fig. 7C). The Paleoproterozoic Strelley Pool Formation records fractionated $\delta^{112}\text{Cd}_{\text{auth}}$ signatures above the BSE background. Yet, at low sample size and absent covariation with Cd/Zn ratios we cannot confidently link these signatures to the evolution of CA and its use of Cd as metallic co-factor. In contrast, the White Mfiozi inlier from the Mesoarchean Pongola Group (S. Africa) shows a clear positive correlation between $\delta^{112}\text{Cd}_{\text{auth}}$ and 1/Cd at elevated Cd/Zn ratios, arguing for Rayleigh-type fractionation⁵⁸ and the use of Cd as a metallic co-factor by phototrophic metabolism (Fig. 7C) in what has been described as early Archean oxygen oasis (whiffs of O_2 in Fig. 7A). Clear Rayleigh-type Cd fractionation patterns are reported here from the Neoproterozoic lacustrine Tumbiana Fm. that overlap in $\delta^{112}\text{Cd}_{\text{auth}}$ and Cd/Zn ratios with modern high productive surface oceans, arguing for an even more definitive biological uptake of Cd in enzymes like CA. Similar patterns have been reported from conophyton-type stromatolitic carbonates of the Neoproterozoic Fazenda Funil Fm. (Fig. 7B) with Rayleigh-type Cd uptake and recycling

under extreme Cd starvation, causing both positive and negative Cd isotopic compositions, coinciding with the redox gradient of the ambient microbial environment⁸⁷. Stromatolitic carbonates of the Miocene Ritzing Fm. support this inference by showing that variable Cd isotope fractionation is linked to redox and nutrient regime shifts during the Badenian Salinity crisis⁸⁸. The essentially modern signature in the Holocene Hamelin Pool stromatolites serves as our calibration point. Together, these snapshots suggest that the enzymatic processes capable of fractionating Cd isotopes (like the uptake of Cd as co-factor into CA) were operational at least by the Meso- to Neoarchean and have persisted in microbial ecosystems since.

Conclusion

The use of carbonic anhydrase (CA) by phototrophs, makes it a crucial enzyme in multiple carbon fixation pathways, including the Calvin-Benson Cycle. The presence of CA implies its central role in facilitating inorganic carbon interconversion and maintaining pH homeostasis and thus accounts for one of the most crucial innovations in microbial metabolism. Modern phototrophic primary producers actively use Zn and Cd as metal co-factors in a cambialistic version of CA. We show, that the biogeochemical cycling of Cd in microbial communities of Holocene Shark Bay, via uptake into CA, is reflected in a large positive fractionation of Cd isotopes recorded in stromatolitic carbonates. Furthermore, our high-resolution trace-metal, Cd, and C isotope compositions support the use of Cd as a metallic co-factor for the operation of CA and related metabolic pathways in microbial habitats from at least the Mesoarchean onward. Although fossil and molecular evidence show that microbial diversity and metabolic complexity were present during the Paleoarchean, metal co-factor usage and carbon metabolism, known as the metallome, were likely more primitive, reflecting early stages of evolutionary development when dissolved trace metals were scarce in microbial habitats. While the Strelley Pool Formation stromatolites lack clear evidence, shifts in trace-metal availability and more substantial Cd isotopic fractionation recorded in stromatolitic carbonates suggest that metallic co-factor uptake into CA and the subsequent fueling of innovative microbial metabolism, such as phototrophy and bacterial sulfate reduction, persisted from the Mesoarchean to the Neoarchean and beyond. To further corroborate the use of CA by early microbial life, future studies should aim for combined analyses of Cd-Zn-C isotopic compositions in a larger set of stromatolitic carbonates to infer on deep time carbon cycling.

Methods

Powdered samples from all three locations were obtained by micro-drilling individual stromatolitic carbonate layers with a diamond-coated, hardened-steel dental drill bit. The small sample number for the Strelley Pool Formation reflects the combined constraints of restricted sampling access to the protected geo-heritage site and the sample mass requirements (~1 g) for high-precision Cd isotope analysis. We here sampled a loose hand specimen collected outside the protected area³³.

For C isotopic compositions, ~100 µg of powders were reacted with 30 µL ~100 % phosphoric acid at 70 °C in a Gasbench II connected to a Thermo Fisher Scientific MAT-253 mass-spectrometer at the State Key Laboratory of Mineral Deposits research (Nanjing, PR China). As a reference gas, CO₂ calibrated against Vienna Pee Dee Belemnite (V-PDB) was used and all compositions are reported as ‰ deviation in the δ notation. Based on the reproducibility of our in-house CaCO₃ reference material GBW04405, the analytical precision is better than 0.1‰ (1 SD).

For trace-element and Cd isotopic analyses ca 1 g of drilled stromatolitic carbonates and reference materials was washed with DI water in acid-washed 50 mL centrifuge tubes to remove the easily leachable fraction and subsequently reacted with pH 5.5 buffered 1 M glacial acetic acid (trace-element grade purity) to dissolve carbonate minerals and leave other authigenic and detrital minerals untouched (e.g., phosphates, sulfides, silicates and oxides)⁸⁹. The leachates were then centrifuged, and the supernatants pipetted into pre-cleaned 50 mL PFA beakers. After drying down on a hotplate, the samples were re-equilibrated with weak double-distilled nitric

acid and aliquots diluted to 1:2000 for trace-element analyses using an Agilent 7900 ICP-MS with internal Rh doping and external calibration to a ten-point multi-element solution at the State Key Laboratory of Marine Geology at Tongji University, Shanghai (SKLMG). Additionally, leachates of three carbonate reference materials (JDo-1, JLS-1 and CAL-S) were analyzed, with their concentrations matching published values using similar leaching methodologies within 5–10% RSD. We further processed alongside a spatially associated host rock in the TF comprising a basaltic tuff (via conc. HNO₃:HCl:HF high-pressure digestion) and a basal substrate carrying one of the SB stromatolitic domes made up of a calcitic mollusk-rich bound stone via acetic acid leaching as described above. Cadmium concentration enrichment factors relative to Bulk Silicate Earth (Cd = ~0.04 µg/g, Al = 23500 µg/g) were calculated using Eq. 1:

$$EF_{Cd} = (Cd_{carb}/Al_{carb})/(Cd_{BSE}/Al_{BSE}). \quad (1)$$

For the calculation of enrichment factors (Eq. 1) and the subsequent derivation of authigenic Cd concentrations and isotopic compositions (Eqs. 2–5), a detrital correction is essential. We selected the BSE composition as our detrital reference standard. This choice is motivated by the temporal scope of our study, which spans the Archean to Holocene. The BSE provides a more consistent and conservative baseline for the integrated detrital flux over deep time, as the composition of the Upper Continental Crust (UCC) is known to have evolved. For transparency and comparison, the UCC baseline is also plotted in all relevant figures. Moreover, the magnitude of the Cd enrichments and isotopic excursions in our samples far exceeds the compositional range defined by either the BSE or UCC, confirming that the calculated authigenic signals are robust and resolvable once detrital-contaminated samples are excluded (Fig. 2).

Aliquots containing at least 5 ng Cd were spiked with an in-house ¹⁰⁶-¹⁰⁸Cd double spike (DS) at a ratio of ~2:1 (sample-DS) and then processed via column purification^{73,90,91}. Cadmium isotopic compositions were analyzed at Freie Universität Berlin (FUB) using a NEOMA MC-ICP-MS using an Apex Omega introduction system. We simultaneously obtained masses of ¹⁰⁶Cd to ¹¹⁶Cd as well as potential interfering elements that were monitored on masses ¹⁰⁵Pd, ¹¹⁵In, and ¹¹⁸Sn and corrected within the MC software. We used a 60 µL/min PFA nebulizer and combined Jet (sample) and X (skimmer) cones. Signals were detected in low-resolution mode using 8-s integration times for 50 individual cycles per sample. When possible, we analyzed each sample twice. We analyzed NIST SRM 3108 as a zero-reference material and corrected for any daily offset by bracketing all measurements to the JMC-Cd standard. We report all Cd isotopic compositions in the δ notation relative to NIST SRM 3108 and obtained δ¹¹²Cd of the geo- standards of CAL-S, JLS-1, and JDo-1, yielding 0.11 ± 0.03‰, 0.13 ± 0.06‰, 0.02 ± 0.05‰ (all 2SE). Our JDo-1 analyses well overlap with literature data⁶⁶. The general precision of our method is in the order of 0.03‰ (2SE) based on repeated analyses of DS-NIST mixtures at a total procedural blank of <40 pg Cd. Authigenic isotopic Cd compositions (subscript auth) were calculated relative to UCC using the following equations:

$$\delta^{112}Cd_{auth} = (\delta^{112}Cd_{carb} - fCd_{det} * \delta^{112}Cd_{BSE})/(1 - fCd_{det}) \quad (2)$$

$$Cd_{auth} = (Cd_{carb} - (Cd_{BSE} * (Al_{carb}/Al_{BSE}))) \quad (3)$$

$$Cd_{det} = Cd_{carb} - Cd_{auth} \quad (4)$$

$$fCd_{det} = (Cd_{det})/Cd_{carb} \quad (5)$$

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All geochemical data conducted for this study are attached in a supplementary information table, as well as in a publicly accessible data repository under: Hohl, Simon (2025), “Carbonic anhydrase traced by Cd isotope fractionations in Archean to Holocene stromatolitic carbonates V.2”, Mendeley Data, V2, <https://doi.org/10.17632/cg3gk2yfvz.2>.

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Author contributions

S.V.H. designed and led the research, acquired funding, performed experiments and wrote the original draft for this manuscript. S.V., A.N.M. and M.V.K. conducted fieldwork in Western Australia discussed the data and revised the manuscript. P.G. provided laboratory resources, helped with data acquisition and revised the final manuscript.

Competing interests

The authors declare no competing interests, either financial or non-financial, that could be perceived as influencing the research, analysis, or interpretation presented in this manuscript.

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During the preparation of this work, the authors used Grammarly in order to improve grammar, syntax, and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication, vouching for the integrity and originality of the final text.

Additional information

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